

CONTINUITY

and DIFFERENTIABILITY

CONTINUITY:-

Let $f: D \rightarrow \mathbb{R}$ be a function and ' a ' is an interior point of D . Then function f is said to be continuous at $x=a$, If for given $\epsilon > 0$, $\exists \delta > 0$ s.t.

$$|x-a| < \delta \Rightarrow |f(x)-f(a)| < \epsilon$$

i.e. $|f(x)-f(a)| < \epsilon$ whenever $x \in (a-\delta, a+\delta) \cap D$

Let $f: D \rightarrow \mathbb{R}$ be a function and ' a ' is an interior point of D . Then f is said to be continuous at $x=a$ if

- (i) $\lim_{x \rightarrow a^-} f(x)$ exists finitely and uniquely
- (ii) $\lim_{x \rightarrow a^+} f(x)$ exists finitely and uniquely
- (iii) $\lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x)$
- (iv) $f(a)$ exists finitely
- (v) $f(a) = \lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x)$

Then f is said to be continuous at $x=a$.

Continuity in (a,b) : $\rightarrow$ If a function f is continuous at each point of (a,b)

Continuity in $[a,b]$: $\rightarrow$ If $f: D \rightarrow \mathbb{R}$ is said to be continuous in $[a,b] \subseteq D$ if

- (i) f is cts in (a,b)
- (ii) $\lim_{x \rightarrow a^+} f(x) = f(a)$
- (iii) $\lim_{x \rightarrow b^-} f(x) = f(b)$

Example of continuous function: $\rightarrow$

- (1) Polynomial function are continuous $\forall x \in \mathbb{R}$
- (11) $|x|$, e^x , $\sin x$, $\cos x$ is continuous $\forall x \in \mathbb{R}$
- (3) Rational function $f(x) = \frac{p(x)}{q(x)}$ is continuous
 $\forall x \in \mathbb{R} - \{x: q(x) = 0\}$
- (4) Irrational function $f(x) = x^{\frac{1}{2n}}$; $n \in \mathbb{N}$
is continuous $\forall x \in [0, \infty)$
- (5) Irrational function $f(x) = x^{\frac{1}{2n+1}}$; $n \in \mathbb{N}$
is continuous $\forall x \in \mathbb{R}$
- (6) $\log x$, $\sec x$, $\tan x$, $\operatorname{cosec} x$, $\cot x$ are continuous in their domains
- (7) $[x]$, $\lceil x \rceil$, $\{x\}$ is discontinuous $\forall x \in \mathbb{Z}$
 $\downarrow$
 $x \leq [x] < x+1$

Algebra of continuous function: $\rightarrow$

- (i) If f and g are continuous at $x=a$
 $\Rightarrow f+g, f-g, f \cdot g$ are continuous at $x=a$
 and $\frac{f}{g}$ is continuous provided $g(a) \neq 0$
- (ii) If f is continuous at $x=a$ and g is continuous at $f(a)$
 $\Rightarrow (g \circ f)(x)$ is continuous at $x=a$
- (iii) If f is continuous at $x=a$ Then $f, f^2, \dots$ are continuous.
 But converse need not be true.
 e.g. $f(x) = \begin{cases} 1 & ; x \in \mathbb{Q} \\ -1 & ; x \in \mathbb{Q}^c \end{cases}$

$\Rightarrow f^2(x) = 1 \quad \forall x \in \mathbb{R}$ which is continuous
 but $f(x)$ is not continuous

- (iv) Composition of two continuous function is continuous but converse may not be true
 e.g. $\sin x^2$ and $\sin^2 x$ are continuous
 $f(x) = \sin x, \quad g(x) = x^2$

$$g \circ f(x) = g(f(x)) = g(\sin x) = \sin^2 x$$

$$f \circ g(x) = f(g(x)) = f(x^2) = \sin x^2$$

- (v) $\max\{f, g\}$ and $\min\{f, g\}$ are continuous if f and g are continuous.

$$\max\{f, g\} = \frac{(f+g) + |f-g|}{2}$$

$$\min\{f, g\} = \frac{(f+g) - |f-g|}{2}$$

(vii) If $g \circ f$ is continuous $\nRightarrow$ f and g are continuous

e.g.

$$f(x) = \begin{cases} -1, & x \in \mathbb{R} \\ 1, & x \in \mathbb{Q}^c \end{cases}; \quad g(x) = |x|$$

$$g \circ f(x) = 1 \quad \forall x \in \mathbb{R}$$

which is continuous

Types of Discontinuity

① Removable discontinuity

Isolated point dis.
 $(\text{LHL} = \text{RHL} \neq f(a))$

Missing point

$(f(a) \text{ विद्यमान नहीं होगा})$

② First type discontinuity

$(\text{LHL} \neq \text{RHL})$
 but both exist finitely

③ Second type discontinuity

LHL or RHL or both DNE finitely

① Removable discontinuity $\rightarrow$ A function $f(x)$ is removable discontinuity at $x = a$ if $\lim_{x \rightarrow a} f(x)$ exists finitely and uniquely, but either $f(a) \neq \lim_{x \rightarrow a} f(x)$ or $f(a)$ is not defined.

e.g.

$$f(x) = \begin{cases} \frac{1 - \cos x}{x^2}; & x \neq 0 \\ 3, & x = 0 \end{cases}$$

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} \quad \left| \frac{0}{0} \text{ form} \right.$$

$$= \lim_{x \rightarrow 0} \frac{\sin x}{2x} = \frac{1}{2} \neq f(0)$$

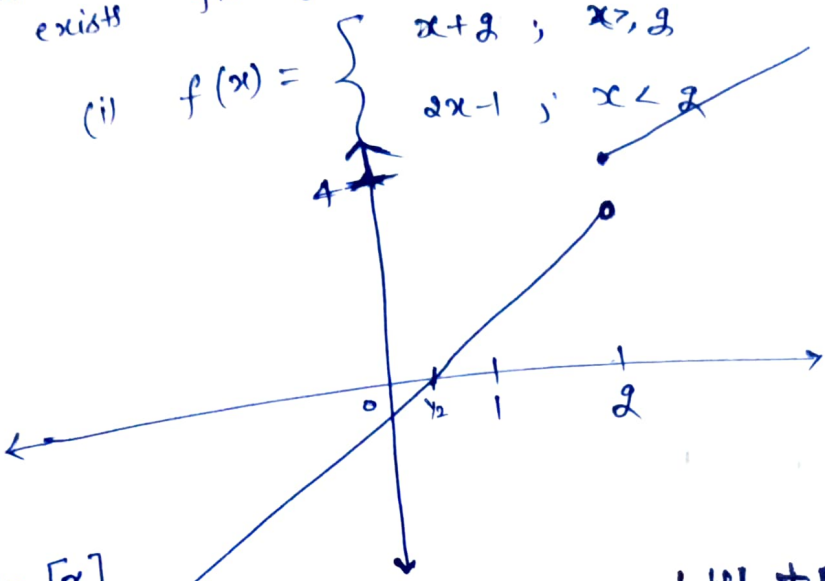
e.g.

$$f(x) = \frac{1 - \cos x}{x^2} ; x \neq 0$$

② First kind discontinuity $\rightarrow$ A function $f(x)$ has first kind discontinuity at $x=a$ if $\lim_{x \rightarrow a^+} f(x)$ and $\lim_{x \rightarrow a^-} f(x)$ exist finitely but not equal.

e.g.

(i) $f(x) = \begin{cases} x+2, & x \geq 2 \\ 2x-1, & x < 2 \end{cases}$



(ii) $f(x) = [x]$

LHL $\neq$ RHL

③ Second kind discontinuity $\rightarrow$ A function $f(x)$ has second kind discontinuity at $x=a$ if either $\lim_{x \rightarrow a^+} f(x)$ or $\lim_{x \rightarrow a^-} f(x)$ or both DNE finitely and uniquely.

e.g. (i) $f(x) = \begin{cases} \frac{1}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$ $\rightarrow$ Infinite discontinuity

(ii) $f(x) = \begin{cases} \sin \frac{1}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$ $\rightarrow$ oscillatory discontinuity

(iii) $f(x) = \begin{cases} 1, & x \in \mathbb{Q} \\ -1, & x \in \mathbb{Q}^c \end{cases}$

DIFFERENTIABILITY

$$\text{Slope of } PQ = \frac{f(c-h) - f(c)}{c-h-c}$$

$$\text{Slope of } PR = \frac{f(c+h) - f(c)}{c+h-c}$$

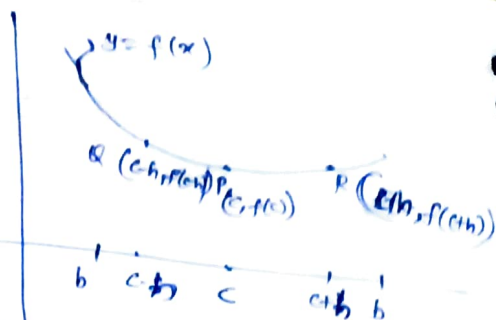
As $h \rightarrow 0$

$$\lim_{h \rightarrow 0} \frac{f(c+h) - f(c)}{-h}$$

$$\text{i.e. } \lim_{Q \rightarrow P} \frac{f(c-h) - f(c)}{-h}$$

$$\text{As } h \rightarrow 0 \quad \text{i.e. Slope of } PR = \text{tangent}$$

$$\lim_{h \rightarrow 0} \frac{f(c+h) - f(c)}{h}$$



$f(x)$ is differentiable at $x=c$

$$\Leftrightarrow \lim_{h \rightarrow 0} \frac{f(c-h) - f(c)}{-h} = \lim_{h \rightarrow 0} \frac{f(c+h) - f(c)}{h}$$

$\Leftrightarrow$ Unique tangent at P

Geometrically: No sharp Edge

Definition: $\rightarrow$

A function $f(x)$ is said to be differentiable at $x=c$ if $\lim_{x \rightarrow c} \frac{f(x) - f(c)}{x-c}$ exists finitely and uniquely and value of this limit is called derivative of $f(x)$ and is denoted by $f'(x)$.

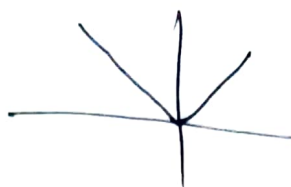
Que:

$$f(x) = |x|$$

check differentiability at $x=0$

$$\begin{aligned} \text{LHD} \quad & \lim_{x \rightarrow 0^-} \frac{f(x) - f(0)}{x-0} \\ &= \lim_{h \rightarrow 0} \frac{f(-h)}{-h} = \lim_{h \rightarrow 0} \frac{|-h|}{-h} = -1 \end{aligned}$$

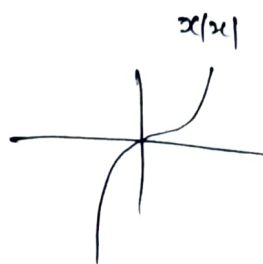
RHD $\lim_{x \rightarrow 0^+} \frac{f(x) - f(0)}{x - 0}$
 $= \lim_{h \rightarrow 0} \frac{|h|}{h} = 1$



LHD $\neq$ RHD
 $\Rightarrow f(x)$ is not differentiable at $x=0$

Que: $f(x) = x|x|$ at $x=0$

$$f(x) = \begin{cases} x^2; & x > 0 \\ -x^2; & x < 0 \end{cases}$$



LHD $\lim_{x \rightarrow 0^-} f(x) = \lim_{h \rightarrow 0} \frac{f(-h)^2}{-h} = 0$

RHD $\lim_{x \rightarrow 0^+} f(x) = \lim_{h \rightarrow 0} \frac{h^2}{h} = 0$

** If $f(x)$ is differentiable at $x=a$ Then $f(x)$ is continuous at $x=a$ but converse need not be true

Let $f(x)$ is differentiable at $x=a$

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} \text{ exists finitely and uniquely.}$$

Now,

$$\lim_{x \rightarrow a} [f(x) - f(a)] = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} \times (x - a)$$

$$= \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} \times \lim_{x \rightarrow a} (x - a)$$

$$\lim_{x \rightarrow a} [f(x) - f(a)] = f'(a) \times 0 = 0$$

$$\Rightarrow \boxed{\lim_{x \rightarrow a} f(x) = f(a)}$$

$\therefore f(x)$ is continuous at $x=a$.